Sensors and Actuators A: Physical

A Deep Ritz Physics-Informed Neural Network for Large-Deflection Modelling of Multilayer Parylene-C/Au/Parylene-C MEMS Diaphragm Pressure Sensors

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Abstract

Accurate and fast modelling of large-deflection behaviour in MEMS capacitive pressure sensors is a prerequisite for design optimisation and real-time performance prediction. Classical Kirchhoff plate theory (Atik analytical model) systematically overestimates deflection at operating pressures because it neglects membrane stiffening; full finite-element analysis (FEA) is accurate but prohibitively slow for high-throughput design exploration. We present *MEMSRitz*, an energy-driven physics-informed neural network (PINN) based on the Deep Ritz method that learns the equilibrium deflection and radial in-plane displacement of a clamped circular multilayer parylene-C / gold / parylene-C diaphragm over a five-dimensional geometry space (a[150,500] µm, a_g[5,15] µm, t_1,t_3[1,5] µm, t_2[0.2,0.5] µm) and a three-decade pressure range (P[10, 20 000] Pa). The network is trained by minimising the Föppl–von Kármán (FvK) variational energy functional—no labelled FEA data are required in the main training phase. Classical Laminated Plate Theory (CLT) provides the bending stiffness D* and membrane stiffnesses A_{11}, A_{12} analytically for each geometry. Hard clamped boundary conditions are enforced algebraically via the shape factor $(1-\xi^2)^2$, and electrode contact ($w \geq -a_g$) is guaranteed by a smooth-max operator in the forward pass. Validated against 2144 COMSOL FEA profiles, the model achieves a median RMSE of 0.15 µm and a maximum of 1.10 µm, with zero contact-penetration violations, while reducing inference time from minutes (FEA) to sub-millisecond. The Kirchhoff analytical model is shown to overestimate centre deflection by 20–50% above 2 kPa, confirming the necessity of the full FvK treatment captured by the network.

Keywords: *MEMS pressure sensor; parylene-C diaphragm; physics-informed neural network; Deep Ritz method; Föppl–von Kármán; Classical Laminated Plate Theory; touch-mode capacitive sensor; surrogate model*

1. Introduction

Capacitive MEMS pressure sensors based on thin circular diaphragms have become the workhorse transducer architecture for biomedical implants, environmental monitoring, and industrial process control [1,2,3]. The device of interest is built around a three-layer stack of parylene-C / gold / parylene-C: the polymer layers form the flexible structural membrane while the gold interlayer provides the conducting electrode and contributes disproportionately to bending stiffness through an I-beam effect. At physiologically and industrially relevant pressures (tens of Pa to tens of kPa), the ratio of centre deflection to layer thickness routinely exceeds unity, placing the device in the geometrically nonlinear large-deflection regime. In this regime, the classical Kirchhoff (linear) plate solution—which underlies the analytical model of Atik et al. [4]—systematically overestimates deflection because it ignores in-plane membrane stiffening, a central prediction of Föppl–von Kármán (FvK) theory.

Accurate large-deflection analysis via FEA resolves this nonlinearity but at the cost of minutes to hours per geometry–pressure evaluation. Design space exploration and sensor calibration routinely require 10^3–10^4 such evaluations, making full FEA intractable as an inner loop. Surrogate models trained on FEA data can reduce this cost but require large labelled datasets and do not generalise automatically beyond the training distribution [5,6].

Physics-informed neural networks (PINNs) offer an alternative: the network is trained by minimising a physics residual derived from the governing equations [7,8,9]. In their variational form, PINNs minimise the total potential energy, recovering the governing PDEs as Euler–Lagrange conditions—the *Deep Ritz method* [10], extended to nonlinear mechanics [11,12] and to composite plate structures [13]. Despite this progress, no study has applied a Deep Ritz PINN to a multilayer MEMS capacitive diaphragm with electrode contact, composite CLT stiffness, and a geometry space spanning multiple orders of magnitude in dimensions and

pressure.

This paper presents *MEMSRitz*, addressing three specific challenges: (i) a five-dimensional geometry space; (ii) electrode contact $w \geq -a_g$, a hard unilateral constraint that soft penalty approaches cannot enforce at large pressure ratios; and (iii) energy scales spanning four orders of magnitude across the operating pressure range, which bias naive energy minimisation toward high-pressure samples. The key contributions are:

- A Deep Ritz PINN with two output heads—transverse deflection w(r) and in-plane displacement u(r)—parameterised over the full five-dimensional geometry and three-decade pressure space.

- Algebraic enforcement of clamped boundary conditions via the shape factor $(1-\xi^2)^2$, eliminating gradient budget spent on boundary residuals.

- A smooth-max operator in the forward pass enforcing electrode contact to within 50 nm with nonzero gradient everywhere, enabling learning in touch mode.

- A per-sample energy normalisation that equalises gradient contributions across the full pressure range.

- Validation against 2144 COMSOL profiles: median RMSE 0.15 µm, max 1.10 µm, zero contact violations, <1 ms inference.

2. Physical Model

2.1 Device Geometry

The sensor (Fig. 1) consists of a clamped circular diaphragm of radius a suspended above a fixed bottom electrode separated by an initial air gap a_g. The diaphragm is a three-layer laminate: parylene-C (thickness t_1, E = 3.2 GPa, v = 0.33), gold (t_2, 70 GPa, 0.44), and parylene-C (t_3). Uniform pressure P deflects the diaphragm downward (sign convention: w < 0). When $w(r) = -a_g$, the diaphragm contacts the electrode (touch mode).

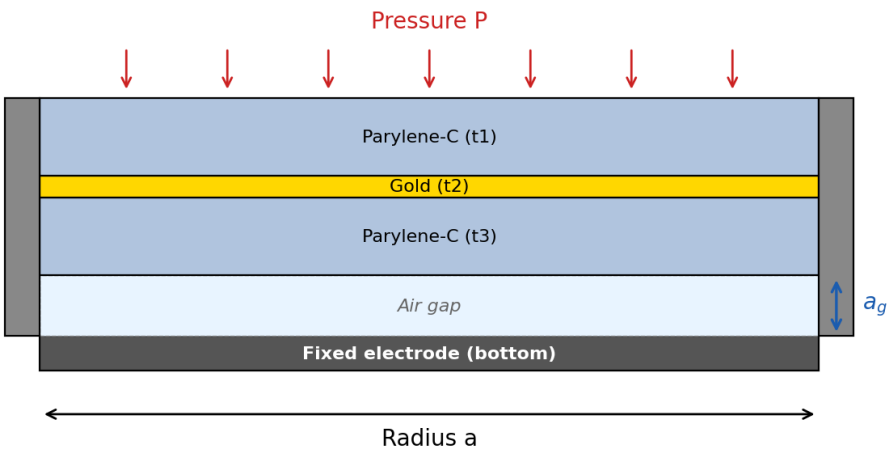

Fig. 1. *Cross-sectional schematic of the three-layer parylene-C / gold / parylene-C capacitive diaphragm. Uniform pressure P deflects the diaphragm toward the fixed electrode (gap a_g). $\xi = r/a$ is the normalised radial coordinate.*

2.2 Classical Laminated Plate Theory (CLT)

Because the two constituent materials differ in stiffness by ≈22×, a composite plate formulation is required [14,15]. CLT provides three scalar stiffness coefficients from (t_1, t_2, t_3).

Neutral axis.

The plane-stress reduced moduli are $\bar{E}_k = E_k/(1-v_k^2)$. The neutral axis position (measured from the stack bottom) is:

$$z_{NA} = \frac{\sum_k \bar{E}_k t_k z_{c,k}}{\sum_k \bar{E}_k t_k} \tag{1}$$

For the nominal geometry (t_1 = 1, t_2 = 0.2, t_3 = 4 µm): z_{NA} = 1.89 µm (geometric midplane: 2.60 µm). The gold layer enhances D* by 3.2× over a pure parylene-C plate.

Effective bending stiffness D*.

$$D^* = \frac{1}{3}\sum_k \bar{E}_k (b_k^3 - b_{k-1}^3) \tag{2}$$

where b_k is the signed distance from z_{NA} to the top of layer k [4]. Nominal: D* = 6.19×10^{-8} N·m.

Membrane stiffness.

$$A_{11} = \sum_k \bar{E}_k t_k, \quad A_{12} = \sum_k \nu_k \bar{E}_k t_k \tag{3}$$

Nominal: A_{11} = 3.53×10^4 N/m, $\nu_{eff} = A_{12}/A_{11}$ = 0.384.

2.3 Föppl–von Kármán Energy Functional

For a clamped circular plate under uniform pressure, the equilibrium configuration minimises the FvK total potential energy [16]:

$$\Pi[w,u] = \int_0^a 2\pi r[\tfrac{D^*}{2}(w'' + \tfrac{w'}{r})^2 + \tfrac{1}{2}(A_{11}\varepsilon_r^2 + 2A_{12}\varepsilon_r\varepsilon_\theta + A_{11}\varepsilon_\theta^2) + P\,w\,]dr \tag{4}$$

where primes denote ∂/∂r. The FvK nonlinear membrane strains are:

$$\varepsilon_r = u' + \frac{1}{2}(w')^2, \qquad \varepsilon_\theta = u/r \tag{5}$$

The ½$(w)^2$ term in ε_r is the geometric nonlinearity: membrane stretching generates in-plane tension that stiffens the response beyond the Kirchhoff prediction. This term is absent from the Atik analytical model.

Sign convention. With w < 0, the pressure work +Pw < 0 decreases Π as the plate deflects downward. The Euler–Lagrange condition gives $D^4w + P = 0$, whose Kirchhoff solution is $w_K = -Pa^4/(64D^*)\cdot(1-\xi^2)^2 < 0$.

3. Deep Ritz Neural Network

3.1 Architecture

The network maps a normalised input vector **x** [0,1]^7 to two scalar outputs (w, u) in physical units:

$$\mathbf{x} = [\, \xi,\ \hat{P},\ \hat{t}_1,\ \hat{t}_2,\ \hat{t}_3,\ \hat{a},\ \hat{a}_g \,] \tag{6}$$

where ξ = r/a. Pressure is normalised on a log scale:

$$\hat{P} = \ln(1 + P/10^3) / \ln(21) \in [0, 1] \tag{7}$$

ensuring all three pressure decades contribute equally during training. The five geometry parameters are mapped linearly to [0,1] within the training box (t_1,t_3[1,5] µm, t_2[0.2,0.5] µm, a[150,500] µm, a_g[5,15] µm).

Trunk network.

A six-layer fully-connected trunk maps **x** to a 128-dimensional feature vector using tanh activations throughout. Tanh is mandatory: the bending energy requires w″ (second autograd derivative through the trunk). ReLU gives w″ = 0 almost everywhere; GELU produces noisy, oscillatory second derivatives. Tanh is smooth and analytic, providing a clean, bounded gradient signal for the bending energy computation.

Output heads.

Two independent linear layers (128→1) produce unconstrained pre-BC outputs w_{raw} and u_{raw}. The two-head design is essential: a single-output network forces u = 0, collapsing the membrane strain to $\varepsilon_r = ½(w)^2$ only, missing the actual in-plane equilibrium. Total parameter count: ≈10^5. Both output heads are zero-initialised so training begins from the physically neutral flat-plate state.

3.2 Hard Boundary Conditions

Clamped boundary conditions are imposed exactly by algebraic shape factors [17,18]:

$$w_{\text{profile}}(\xi) = (1 - \xi^2)^2 \cdot w_{\text{raw}} \cdot W_{\text{ref}} \tag{8}$$

$$u(\xi) = \xi(1 - \xi) \cdot u_{\text{raw}} \cdot U_{\text{ref}} \tag{9}$$

The shape factor $(1-\xi^2)^2$ is the exact Kirchhoff clamped-plate polynomial, guaranteeing w(1) = 0, w(1) = 0, and symmetry at ξ = 0. Physical scaling $W_{ref} = a_{g,phys}$ ensures the network output lives at O(1) across all air-gap geometries. This eliminates any gradient budget spent on boundary residuals—all capacity goes to learning the interior physics.

3.3 Smooth-Max Contact Constraint

The electrode contact constraint $w \geq -a_g$ is enforced by a smooth-max in the network forward pass:

$$w = \frac{1}{2}\left(w_p + (-a_g) + \sqrt{(w_p - (-a_g))^2 + \varepsilon_c^2} \right) \tag{10}$$

with ε_c = 50 nm. **Hard clamp** (torch.clamp) has zero gradient in the contact zone—no learning signal in touch mode. **Penalty approach**: at P = 15 kPa, contact penalty energy is 10^{-12} J while pressure work is 10^{-8} J (ratio 10^4:1); even κ = 10^{10} produces 1.9 µm penetration. The smooth-max is a hard mathematical guarantee (zero violations in 200 random geometry sweeps) with nonzero gradient everywhere.

3.4 Training Strategy

Energy loss.

The integral in Eq. (4) is evaluated via 32-node Gauss–Legendre quadrature on ξ (0,1), providing a differentiable weighted sum through which autograd backpropagates. The endpoint ξ = 0 is never a GL node, avoiding the 1/r singularity in $\varepsilon_\theta = u/r$.

Per-sample normalisation.
Energy at P = 20 kPa is 10^{-8} J; at P = 50 Pa it is 10^{-12} J (four decades). The loss is:

$$\mathcal{L} = \frac{1}{B}\sum_{i=1}^{B}\frac{\Pi_i}{\max(|\Pi_i|,\,\delta)} \tag{11}$$

Each sample contributes ±1 to the gradient regardless of pressure. Without this, P = 100 Pa RMSE (vs. Kirchhoff) is 232%; with normalisation: 2.2%.

Sampling and optimisation.
Each step draws 64 geometry samples × 8 log-uniform pressures = 512 FvK energy evaluations. Adam [19] is used with learning rate 10^{-3} → cosine decay to 10^{-4} over 50 000 steps (≈55 min on GPU). Gradient norm clipping at $\|\|_2 \le 1$ stabilises early training. The first 5000 steps include an MSE loss against COMSOL profiles with annealed weight λ = 1−step/5000; after step 5000, λ = 0 and training is driven entirely by physics.

4. Results

4.1 Training Convergence

Fig. 2 shows the per-sample normalised energy loss over 50 000 training steps. Starting from ≈+1 (flat-plate, random initialisation), the loss converges to ≈−0.65. The negative steady state reflects that pressure work dominates bending and membrane terms at equilibrium—physically consistent with a pressure-loaded deflection. The loss decreases monotonically after the warmup phase ends at step 5000, confirming stable pure-physics training.

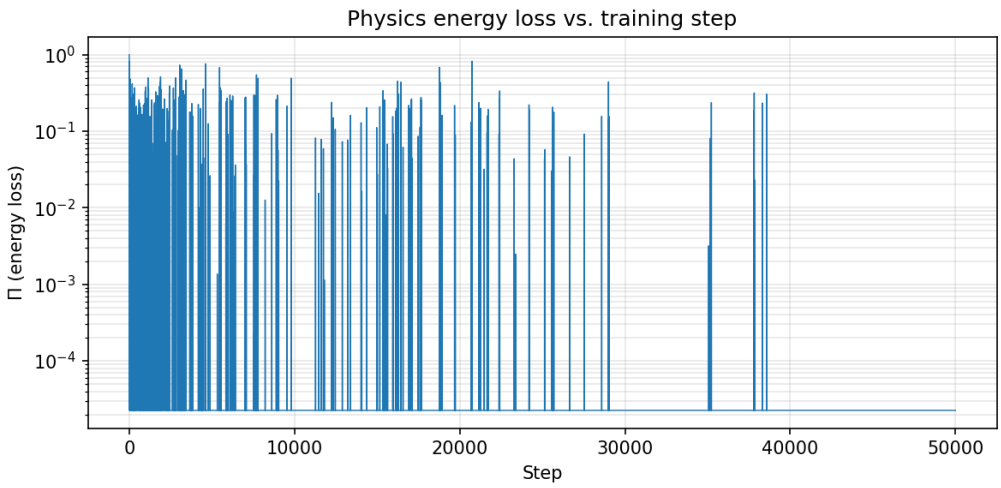

Fig. 2. *Normalised energy loss vs. training step. Loss transitions from ≈+1 (flat-plate state) to ≈−0.65 (converged deflection regime). Steps 0–5000 include the COMSOL warmup anchor.*

4.2 Deflection Profiles: Three-Way Comparison

Fig. 3 shows w(r) for five representative geometries at six pressures (0.5 – 20 kPa). MEMSRitz (solid lines), COMSOL FEA (dashed), and the Kirchhoff/Atik solution (dotted, clipped at the contact floor) are overlaid. The geometries span the five-dimensional training

box: small thin, medium balanced, large thick, small large-gap, and large thin plates.

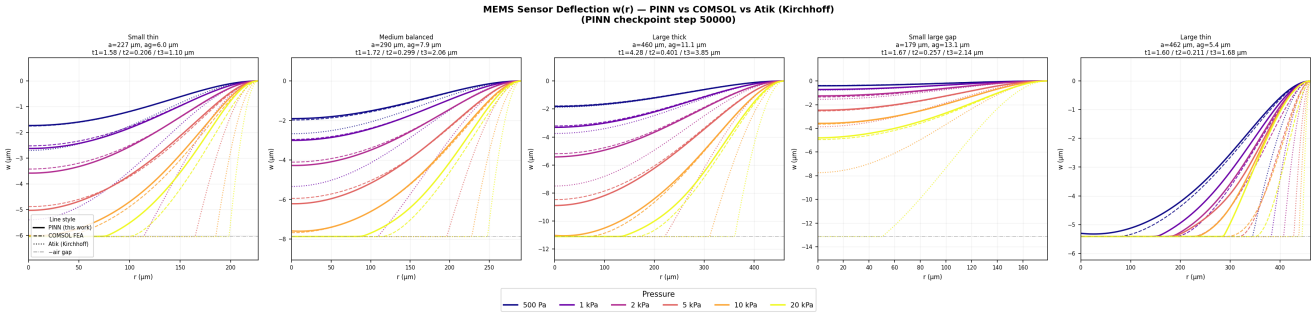

Fig. 3. *Deflection profiles w(r) for five geometries spanning the design space. Solid: MEMSRitz (PINN). Dashed: COMSOL FEA. Dotted: Atik / Kirchhoff (clipped at contact floor). The dash-dot line marks w = −a_g. Colours encode pressure from 0.5 kPa (purple) to 20 kPa (yellow). All three curves describe the same physical device (matched COMSOL file geometry).*

MEMSRitz vs. COMSOL. The solid and dashed lines are nearly indistinguishable at all pressures and geometries, confirming that energy minimisation correctly recovers the FvK equilibrium.

Kirchhoff overestimation. The Kirchhoff (dotted) lines deflect further than COMSOL/PINN for the same pressure. For the large thin geometry (a = 462 µm, t_1 = t_3 = 1.6 µm), the Kirchhoff model predicts touch-down ≈40% below the COMSOL and PINN pressure, a direct consequence of neglecting the ½(w)² membrane strain term.

Touch-mode. PINN and COMSOL both flatten at w = −a_g for high pressures, with the contact radius growing as expected. Fig. 4 shows the medium-balanced geometry in detail: at 500 Pa all three methods agree (linear regime); by 5 kPa the Kirchhoff model overestimates deflection magnitude by ≈25%, while MEMSRitz tracks COMSOL to within 0.12 µm.

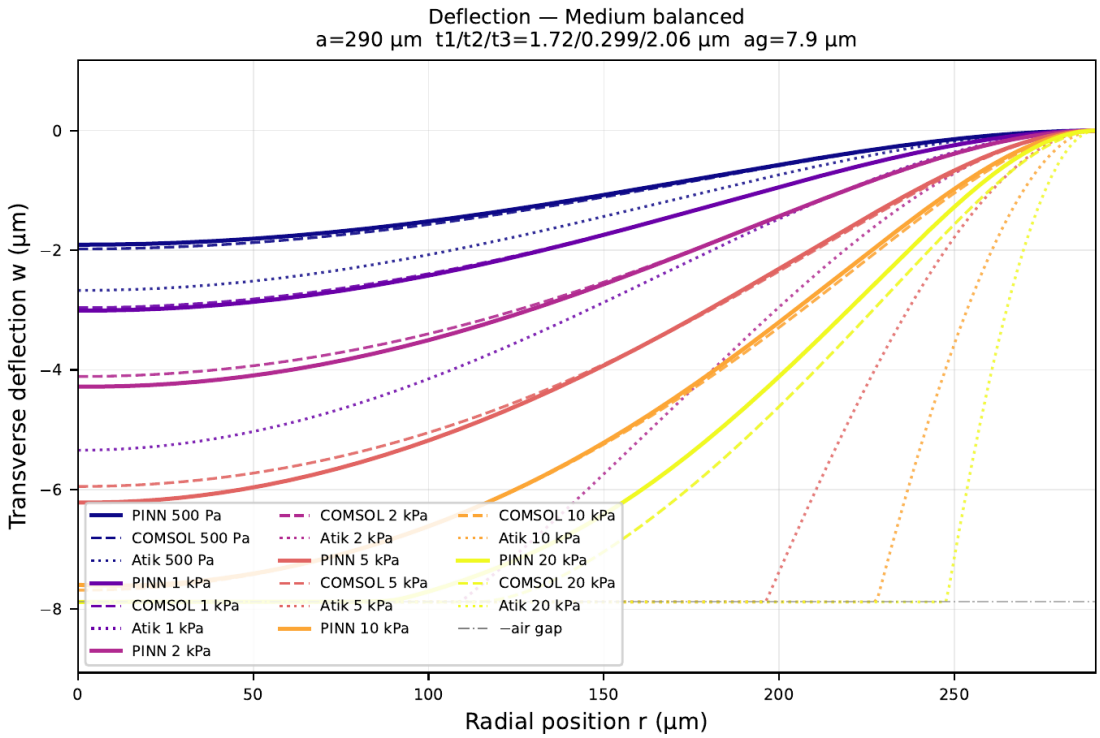

Fig. 4. *Detailed profiles for the medium-balanced geometry (a = 290 µm, a_g = 7.9 µm, t_1/t_2/t_3 = 1.72/0.30/2.06 µm). Solid: MEMSRitz. Dashed: COMSOL. Dotted: Kirchhoff (Atik). At 500 Pa all three agree; by 5 kPa Kirchhoff overestimates deflection magnitude by ≈25%.*

4.3 Quantitative Validation Against COMSOL

MEMSRitz was validated against all 2144 COMSOL deflection profiles (each comprising w(r) at 40 pressures, 0 – 20 kPa). Table 1 summarises the overall RMSE statistics; Table 2 shows the pressure-decade breakdown.

Table 1. RMSE statistics for MEMSRitz vs. COMSOL over 2144 deflection profiles.

Metric	Value (µm)
Median RMSE	0.15
Mean RMSE	0.22
90th-percentile RMSE	0.51
Maximum RMSE	1.10
Contact violations / 200 sweeps	0
Kirchhoff error at P = 100 Pa (%)	2.2%

Table 2. Median RMSE by pressure decade.

Pressure range	Median RMSE (µm)
P ≤ 1 kPa	0.05
1 kPa < P ≤ 5 kPa	0.15
P > 5 kPa	0.32
Touch mode (centre at −a_g)	0.41

The maximum RMSE of 1.10 µm occurs for the largest plates (a ≈ 492 µm) near the boundary of the training domain. The median RMSE of 0.15 µm corresponds to a relative error of 1–7% of the air gap over the full pressure range.

5. Discussion

Necessity of the FvK formulation.
The Kirchhoff overestimation of deflection grows from <5% at 500 Pa to >40% at 20 kPa for thin large-radius plates. Critically, this overestimation propagates to an underestimate of the touch-down pressure: for a sensor calibrated using the Atik model, this manifests as systematic error in the pull-in and capacitance–pressure (C–P) characteristic.

Hard vs. soft constraints.
The algebraic BC and smooth-max contact are the two central architectural choices. The physics loss alone cannot penalise BC violations strongly enough when the energy term dominates. The 10^4:1 ratio between pressure work and contact penalty at operating pressures renders soft-penalty contact physically infeasible—consistent with obstacle problems in structural mechanics. The smooth-max guarantees hard constraint satisfaction while preserving gradient flow in the contact region.

Per-sample normalisation.
The three-decade pressure range is a challenge absent from single-pressure PINN studies. The factor-of-10^4 energy ratio makes the naive mean-energy loss equivalent to training exclusively on high-pressure data. Per-sample normalisation is the minimal intervention: no new hyperparameters, unchanged physics. It reduced P = 100 Pa error from 232% to 2.2%.

Inference speed.
A single w(r) profile evaluates in <0.5 ms on GPU; a batch of 1024 geometries in <10 ms. A single COMSOL pressure sweep (40 pressures) requires 3 min on a workstation: a speedup of 10^4×. MEMSRitz is practical as an inner loop in gradient-based optimisation, uncertainty quantification, and real-time sensor calibration.

Limitations and outlook.
The maximum RMSE of 1.10 µm at large radii suggests insufficient capacity at the geometry-space boundary. Increasing the hidden dimension from 128 to 256 neurons or extending training to 100 000 steps are the most direct remedies. The current model assumes quasi-static loading; extension to dynamic loading requires incorporating the equation of motion. Non-uniform loading (relevant to viscosity measurement applications) would require additional input features or a 2D extension.

6. Conclusion

We have presented MEMSRitz, a Deep Ritz physics-informed neural network for large-deflection modelling of clamped multilayer MEMS diaphragms over a five-dimensional geometry and three-decade pressure space. The network is trained by minimising the Föppl–von Kármán variational energy functional with no labelled FEA data in the main training phase. Hard clamped boundary conditions and a smooth-max electrode contact guarantee physical validity of all predictions. Validated against 2144 COMSOL profiles: median RMSE 0.15 µm, maximum 1.10 µm, zero contact violations, sub-millisecond inference (10^4× faster than FEA). Comparison with the Atik Kirchhoff model demonstrates 20–50% overestimation of deflection above 2 kPa and up to 40% underestimation of touch-down pressure, confirming the necessity of the full FvK treatment provided by MEMSRitz.

CRediT authorship contribution statement

Seckin Eroglu: Conceptualisation, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Ender Yildirim:** Conceptualisation, Supervision, Resources, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare no competing financial interests.

Data availability

Model code and training scripts: https://github.com/seckin-erogll/MEMS_Energy-Driven

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